A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Captive breeding, habitat creation & management - Set 1

Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1 In-situ:** conserving species in their natural habitat with the whole community – ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding **1** Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Is the wild population threatened? **1** Is there a genetically diverse captive population? **1** Is in-situ conservation failing? **1** Is keeping a captive population realistic (habitat size, food, relationships, cost)? **1** Is release likely to be successful now or in future (suitable habitat, threats removed)? **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
01.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Blue whale:** habitat size far too large to provide – food (huge quantities of krill / plankton) cannot be supplied – cost **2 Large blue butterfly:** complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs – requires specific ant species and thyme food plant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Very small gene pool so individuals are closely related and inbreeding is likely **1** Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low **1** A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) **1** Exchange of individuals / semen between zoos – avoid hybridisation with related species / subspecies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided **1 Population interactions:** mixing breeding and non-breeding individuals causes conflict – some need a choice of partners, others pair for life **1 Breeding habitat needed:** courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) **1** Stress / disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1 AI:** semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet – avoids dangerous mating (elephants, rhinos) and partner rejection **1 Embryo transfer:** a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) **1** So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1 Hard release:** individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) **1 Soft release:** gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds **1 Problems faced:** finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Large enough suitable habitat for a viable population **1** Reliable food supply **1** Low predation risk / original threat removed **1** Suitable breeding sites **1** Water **1** Support of the local human population **1** Official / legal protection of habitat and species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	5
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed **1** Identifies causes of failure so future releases can be improved / extra support given **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) **1** The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino **1** Very low success rates with wildlife so far – deep-frozen cells could allow clones even after the last individuals die **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Micro-propagation:** tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant **1 Black poplar:** cuttings / pollarded stems of pure native trees are rooted to produce many identical trees **1** Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild **1** Seeds collected worldwide – target of 10 000 seeds per species to maintain a diverse gene pool **1** Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident **1** Seeds germinated periodically to check viability and replenish stocks – can be used for reintroduction / breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding) – species higher up food chains need larger areas (lower density) **1** Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) **1 Shape:** the perimeter has edge conditions (a mix of neighbouring habitats) – a compact shape maximises core habitat for species needing core conditions **1** Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1 Varied topography:** banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) **1** Islands provide roosting / nesting sites safe from terrestrial predators (foxes) **1** Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet) – scrapes for skylarks – different water depths for waders and ducks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mobile species (birds, flying insects) colonise easily **1** Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced – keystone species especially **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community	2
05.4	1 mark for the point below. **1** Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out – avoids hazards (eg rope bridges for howler monkeys over roads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	
06.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Light:** selective felling to create clearings for plants needing high light while retaining dense canopy for shade species – clearings also give warm sand / soil for lizard eggs – leave dead wood – woodland floor plants (bluebells) grow before the canopy leafs **2 Dissolved oxygen:** narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae – shallow water warms quickly for egg development **2 Nutrients:** removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground – dung patches add local nutrients **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	6

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Food:** wildflower grasslands for seed-eating birds – providing food species **1 Control of predation:** island nest sites unreachable by foxes – trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) **1 Control of competitors:** removing rhododendron so wildflowers get light **1 Pollination:** a range of flowering plants with different flowering times so pollinators have food all season **1** Seed dispersal species (hippos, elephants, monkeys) conserved – control of introduced pathogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession **1** Chalk grassland was produced by centuries of sheep grazing – without it succession to scrub and woodland resumes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
07.2	**1 mark each** 1 mark for each of the 4 points below. **1** Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out **1** Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles **1** Dung creates small nutrient-rich patches supporting different plants and invertebrates – sheep eat short grasses and some bushes **1** Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting – burning removes vegetation but also kills invertebrates and allows scrub regrowth – abandoned plot dominated by scrub (55%) so few species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

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Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sheep eat short grass (some breeds eat scrub) – ponies avoid flowers but eat bracken – cattle remove long rough grass **1** Choice depends on the vegetation to be controlled and the plants to be protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground **1** Heat stimulates dormant heather seeds to germinate so young heather re-establishes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump **1** About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages **1** Each age has its own community: newly cut areas get high light for wildflowers and butterflies – older areas give dense cover for nesting birds / dormice **1** Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour **1** Fewer young trees eaten so trees and riverside vegetation recovered **1** More trees allowed beaver populations to increase – beaver dams increased wetland area **1** Wetland species (birds, fish, invertebrates) became more common – overall biodiversity rose, though wolves cause problems for livestock farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4

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Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species **1** Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	

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